Atlas 126: A Triangle–C_4 Vertex-Amalgam Supporting Plane

Abstract

Let H be a triangle and a 4-cycle sharing exactly one vertex. We prove that every graphon of edge density $p \in [1/2, 1]$ satisfies

$$t(H, W) \geq p^2(2p-1)\big(p^3 + (1-p)^3\big).$$

This is the full chromatic interval and the exact target for Graph Atlas 126. Rooting at the common vertex factors the density into a rooted triangle and a rooted C_4. A two-piece weighted Cauchy–Schwarz projection then reduces the graphon problem to a scalar function of the degree, the one-step degree average, and the rooted triangle density. Its sharp supporting plane uses Reiher's triangle-density theorem on $1/2 \leq p \leq 2/3$ and is elementary above $2/3$. The remaining rational polynomial inequalities are proved by an exact Bernstein subdivision certificate, including exact local analyses of all zero-coefficient boxes.

1 The graph and its chromatic target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. All equalities and inequalities below hold after changing functions on null sets if necessary. Every integrand is nonnegative and bounded, so Tonelli's theorem justifies all changes in the order of integration.

The Atlas-labelled edge set of Graph Atlas 126 is

$$\{02, 05, 13, 14, 23, 34, 35\}.$$

It has graph6 string `ERUO` and degree sequence $(4, 2, 2, 2, 2, 2)$. Its triangle is 1–3–4–1, and its 4-cycle is 0–2–3–5–0; these subgraphs share only vertex 3. Thus it is isomorphic to a triangle and a C_4 with one vertex identified.

If r colours are available, first colour the common vertex. The remaining two triangle vertices have $(r-1)(r-2)$ choices. The three remaining cycle vertices have $(r-1)(r^2-3r+3)$ choices. Hence

$$P_H(r) = r(r-2)(r-1)^2(r^2-3r+3), \qquad \chi(H) = 3. \tag{1}$$

The chromatic target is

$$\begin{aligned}\Phi_H(p) &= (1-p)^6 P_H\left(\frac{1}{1-p}\right) \\ &= p^2(2p-1)\big(p^3 + (1-p)^3\big).\end{aligned} \tag{2}$$

The required interval is therefore $p \in [1/2, 1]$.

2 Rooting and the Hilbert-space projection

Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y),$$

and define

$$a(x) = (T_W d)(x) = \int_{\Omega} W(x,y) d(y)\,\mathrm{d}\mu(y),$$
$$t(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Thus $t(x)$ is the triangle density rooted at x. For fixed x, put

$$b_x(z) = \int_{\Omega} W(x,y)W(y,z)\,\mathrm{d}\mu(y).$$

The C_4 density rooted at x is

$$r_4(x) = \int_{\Omega} b_x(z)^2\,\mathrm{d}\mu(z).$$

Moreover,

$$\int b_x = a(x), \qquad \int W(x,z) b_x(z)\,\mathrm{d}\mu(z) = t(x).$$

Splitting $r_4(x)$ with weights $W(x,z)$ and $1 - W(x,z)$ and applying Cauchy–Schwarz on each part gives, for $0 < d(x) < 1$,

$$r_4(x) \geq \frac{t(x)^2}{d(x)} + \frac{(a(x) - t(x))^2}{1 - d(x)}. \tag{3}$$

Conditioning on the common vertex makes the triangle and C_4 factors independent. Consequently

$$t(H,W) = \int_{\Omega} t(x) r_4(x)\,\mathrm{d}\mu(x) \geq \int_{\Omega} F_p(d(x), a(x), t(x))\,\mathrm{d}\mu(x), \tag{4}$$

where

$$F_p(d,a,t) = \frac{t^3}{d} + \frac{t(a-t)^2}{1-d}. \tag{5}$$

At $d = 0$ and $d = 1$ this is interpreted by the actual rooted limits described in Section 7; no assertion about arbitrary scalar triples at those endpoints is being made.

3 The exact scalar feasible region

The rooted quantities obey

$$0 \leq d \leq 1, \qquad \max\{0, d + p - 1\} \leq a \leq d, \qquad \max\{0, 2a - p\} \leq t \leq \min\{a, d^2\}. \tag{6}$$

Indeed, $a \leq d$ follows from $d(y) \leq 1$. Expanding $\int (1 - W(x,y))(1 - d(y))\,\mathrm{d}\mu(y) \geq 0$ gives $a \geq d + p - 1$. The inequalities $t \leq a$ and $t \leq d^2$ follow by deleting, respectively, one and two edge factors from the rooted triangle integral. Finally,

$$W(x,y)W(x,z) \geq W(x,y) + W(x,z) - 1;$$

multiplication by $W(y,z)$ and integration give $t \geq 2a - p$.

The three global moments needed below are

$$\int d = p, \qquad \int a = \int d^2, \qquad \int t = t(K_3, W). \tag{7}$$

The middle identity follows directly from symmetry and Tonelli: $\int_x a(x) = \int_y d(y)^2$.

For $p \in [1/2, 2/3]$, introduce the unique $x \in [0,1]$ such that

$$p = \frac{3 + 2x - x^2}{6}, \qquad g(p) = \frac{x(3-x)^2}{18}. \tag{8}$$

The K_3 case of Reiher's clique-density theorem gives

$$t(K_3, W) \geq g(p). \tag{9}$$

This is the only prior non-elementary graph-density theorem used here.

4 The supporting planes

Put $q = 1 - p$, $c = 2p - 1$, and

$$C(p) = p^2 c(p^3 + q^3), \qquad D_0 = p^2 + q^2, \qquad h = 1 - \frac{p}{2}.$$

Define the low coefficients

$$\gamma_L = \frac{1}{16} + \frac{c}{2} + c^2, \qquad \rho = \frac{p^2}{D_0}, \qquad \beta_L = 2\gamma_L \rho. \tag{10}$$

On region A, $0 \leq x \leq 13/20$, set

$$\gamma = \gamma_L, \qquad \beta = \beta_L, \qquad \lambda = \frac{C + \beta_L(\rho - p) + \gamma_L(p/2 - \rho^2)}{g} + \frac{x}{1000}. \tag{11}$$

The quotient has a removable singularity at $x = 0$; its prescribed value is $\lambda(0) = 1/16$. The small final term vanishes at the threshold and strictly separates one otherwise isolated boundary contact.

On region C, $13/20 \leq x \leq 1$, use

$$x = \frac{13 + 7w^2}{20}, \qquad 0 \leq w \leq 1, \qquad \varphi = 2w - w^2. \tag{12}$$

Set

$$\begin{aligned} \gamma_H &= 2p(9p^2 - 10p + 3), \\ \beta_H &= p(30p^3 - 29p^2 + 6p + 1), \\ \gamma &= (1 - \varphi)\gamma_L + \varphi\gamma_H, \\ \beta &= (1 - \varphi)\beta_L + \varphi\beta_H, \end{aligned} \tag{13}$$

$$\lambda = \frac{C + \beta(h - p) + \gamma(p/2 - h^2)}{g}. \tag{14}$$

The endpoint-flat interpolation is important at $p = 2/3$. At $w = 1$,

$$\beta = \frac{2}{3}, \qquad \gamma = \frac{4}{9}, \qquad \lambda = 0. \tag{15}$$

On region H, $2/3 \le p \le 1$, use (13) directly and set $\lambda = 0$.

The exact univariate Bernstein checks in the companion verifier prove $\gamma \ge 0$ and $\lambda \ge 0$. They also prove the endpoint scalar gaps used in Section 7. The Bernstein degrees and numbers of zero coefficients for $(\gamma, \lambda, G_0, G_1)$ are

region	γ	λ	G_0	G_1
A	$(4, 0)$	$(16, 0)$	$(14, 0)$	$(17, 0)$
C	$(14, 0)$	$(30, 2)$	$(30, 0)$	$(34, 0)$
H	$(3, 0)$	identically zero	$(5, 0)$	$(5, 2)$

where

$$\begin{aligned} G_0 &= \beta p + \lambda g - C, \\ G_1 &= p^3 - C - \beta(1-p) - \gamma(p-1) - \lambda(p-g). \end{aligned}$$

Lemma 4.1 (Scalar supporting plane)**.** *Let $p \in [1/2, 1)$ and let (d, a, t) satisfy (6) with $0 < d < 1$. With the coefficients from the appropriate region above,*

$$F_p(d, a, t) \ge C(p) + \beta(d - p) + \gamma(a - d^2) + \lambda(t - g), \tag{16}$$

where $g = 0$ and $\lambda = 0$ in region H. The same inequality holds at the actual rooted endpoint triples and at $t = 0$ under the conventions of Section 7.

5 Convex face reduction

Fix d and $t > 0$. After moving the right side of (16) to the left, the part depending on a is

$$\frac{t(a-t)^2}{1-d} - \gamma a.$$

It is convex and has its unconstrained minimum at

$$a_* = t + \frac{\gamma(1-d)}{2t} > t. \tag{17}$$

For fixed d, t, (6) says

$$\max\{t, d - q\} \le a \le \min\left\{d, \frac{t+p}{2}\right\}.$$

Since $a_* > t$, a minimum cannot occur only on the face $a = t$; if that face is a singleton it is already one of the upper faces. It is enough to check

$$a = d - q, \qquad a = d, \qquad a = \frac{t+p}{2}, \qquad a = a_*. \tag{18}$$

For the interior candidate, feasibility of (17) is expressed by

$$\begin{aligned} K_1 &= pt - t^2 - \gamma(1-d) \geq 0, \\ K_2 &= 2t(d-t) - \gamma(1-d) \geq 0, \\ K_3 &= 2t(t-d+q) + \gamma(1-d) \geq 0, \qquad (19) \\ K_4 &= d^2 - t \geq 0. \qquad (20) \end{aligned}$$

The redundant inequality $2t - \gamma \geq 0$ follows from

$$2K_1 + K_3 = (1-d)(2t-\gamma).$$

Substitution of (17) into the residual and multiplication by $4dt > 0$ gives the verifier target

$$4dt\left(\frac{t^3}{d} + \gamma d^2 - \beta(d-p) - C + \lambda g - (\gamma+\lambda)t - \frac{\gamma^2(1-d)}{4t}\right) \geq 0. \qquad (21)$$

Its denominator depends only on the region parameter and has a strictly positive Bernstein expansion. The certificate takes $t = pv$ on the unit cube in the region parameter, d, and v, pruning a box only when one of (19) or $2t - \gamma$ is strictly negative there.

For completeness, the low-density boundary cubes are as follows. Here $u, v \in [0,1]$ and $h = 1 - p/2$:

face	range substitution	t
$a = d - q$, I	$d = q + pu/2$	$(d-q)v$
$a = d - q$, II	$d = h + pu/2$	$2d + p - 2 + (1-d)v$
$a = d$, I	$d = pu/2$	$d^2 v$
$a = d$, II	$d = p/2 + pu/2$	$2d - p + (d^2 - 2d + p)v$
$a = (t+p)/2$, I	$d = p/2 + qu$	$(2d-p)v$
$a = (t+p)/2$, II	$d = h + pu/2$	$2d + p - 2 + 2qv$

These cubes cover every feasible boundary point. Some contain extra infeasible points; positivity there only strengthens the certificate. In region C, the last coordinate on the two $a = (t+p)/2$ faces is split at $v = 1/3$ to isolate the $p = 2/3$ equality.

The high-density region uses coordinates adapted to its curved feasible edges. On $a = d$, after the first cube above, write

$$s = 1 - d, \qquad q = s^2 u, \qquad t = 2d - p + (s^2 - q)v. \qquad (22)$$

The additional constraints $q \leq 1/3$ and $2d - p \geq 0$ give an exact cover. On $a = (t+p)/2$, write

$$q = \frac{X}{3}, \quad p = 1 - q, \quad s = 1 - d, \quad t = p - 2s + 2qv. \qquad (23)$$

The remaining feasibility conditions are exactly

$$s - qv \geq 0, \qquad s^2 + q - 2qv \geq 0. \qquad (24)$$

The rectangle is split at $s = q$ and $v = q$, producing ll, lh, rl, rh. The lower face $a = d - q = p - s$ is similarly split at $s = q$ and uses $t = p - 2s + sv$. These substitutions remove the square-root boundary that would otherwise occur on $a = d$ and expose every endpoint zero in rational coordinates.

6 Exact Bernstein certificate

If a polynomial P on a box is written in the tensor Bernstein basis, then

$$P = \sum_{\boldsymbol{i}} b_{\boldsymbol{i}} \prod_{j} B_{i_j,n_j}(u_j).$$

The basis functions are nonnegative and sum to one. Thus nonnegative $b_{\boldsymbol{i}}$ prove $P \geq 0$ on the box. Exact de Casteljau bisection gives the coefficients on each dyadic child. Conversely, if every coefficient of a feasibility constraint is strictly negative, that box is infeasible. The verifier clears a positive common denominator and performs de Casteljau subdivision over the integers. A separate exact check proves equivalence to rational subdivision up to a positive scale.

After each face substitution, the residual is multiplied by $d(1-d)$; in the interior it is multiplied by $4dt$. All other denominators depend only on the region parameter, and their strict signs are proved by univariate Bernstein coefficients. The certificate tree is:

region	piece	visited	accepted	infeasible
A	interior	1097	261	288
	lower I, II	34	18	0
	upper-d I, II	6	4	0
	upper-t I, II	538	270	0
C	interior	479	106	134
	lower I, II	36	19	0
	upper-d I, II	6	4	0
	upper-t I low/high	7704	3853	0
	upper-t II low/high	7210	3606	0
H	interior	221	42	69
	lower faces	37	20	0
	upper-d faces	2	2	0
	upper-t first range	44	23	0
	upper-t split ll, lh, rl, rh	11670	5836	1
	total	29084	14064	492

Accepted counts leaf boxes proved directly or by an exact local lemma; internal nodes are included in visited.

There are genuine equality boxes, so subdivision alone is not used as a substitute for endpoint proofs. The verifier checks:

- At the region-A interior equality $x = 0$, after $d = 1/2 + y$ and $v = 1/4 + z$, the quadratic part is

  $$5y^2 - 8yz + 8z^2 = y^2 + 4(y - z)^2 + 4z^2.$$

  The higher remainder is dominated on the box of radius $1/64$, while the quotient by x has 380 nonnegative tensor Bernstein coefficients there.

- The isolated region-A upper-face corner has leading terms, after a positive common scale,

  $$112500z + 86021x^2 - 195000xy + 112500y^2.$$

  The quadratic matrix remains positive definite after subtracting $500(x^2 + y^2)$; the exact remainder radius is $1/2048$.

- Each of the four region-C corners at $p = 2/3$ has a positive-definite leading form in $((1-w)^2, u, z)$. Sylvester's criterion is checked over the rationals after subtracting the margins $19/100, 19/100, 3/20, 9/50$; the exact radii are $1/128, 1/256, 1/512, 1/1024$.
- On the high ll piece a weighted quadratic form controls the endpoint uniformly in p. On the high rh piece there are separate exact estimates at $p = 1$ and $p = 2/3$, together with 13 rational dyadic parameter bands. The remaining high upper-face corner has the positive-cone form

$$\frac{119r^2 + 100ru - 88rz + 28u^2 - 16uz + 32z^2}{1458}.$$

  Dropping the nonnegative $100ru$ term leaves a matrix positive definite after subtracting $I/200$, with exact radius $1/128$.

All polynomial construction, coefficient extraction, feasibility pruning, local identities, margins, remainders, tree sizes, graph identification, and target algebra were checked in exact rational arithmetic. There is no floating-point sign decision. This proves Lemma 4.1.

7 Integration and endpoints

Integrating (16) and using (7) gives

$$\begin{aligned}\int F_p(d,a,t) &\geq C(p) + \beta\left(\int d - p\right) + \gamma\left(\int a - \int d^2\right) + \lambda\left(\int t - g\right)\\ &\geq C(p).\end{aligned}$$

The last inequality is (9) and $\lambda \geq 0$ on the low interval; on the high interval the last term is absent. Combining this with (4) proves the desired graphon inequality once the rooted endpoints are interpreted as follows.

If $d(x) = 0$, then $W(x,\cdot) = 0$ almost everywhere, so $a(x) = t(x) = 0$ and the projected contribution is 0. Its scalar gap is $G_0 = \beta p + \lambda g - C \geq 0$. If $d(x) = 1$, then $W(x,\cdot) = 1$ almost everywhere, so $a(x) = t(x) = p$, the rooted C_4 projection is p^2, and $F = p^3$. Its scalar gap is $G_1 \geq 0$. When $t = 0$ and $0 < d < 1$, define $F = 0$; the relevant minima occur on the certified boundary faces. This removes every zero denominator without appealing to continuity outside the actual feasible set.

At $p = 1/2$, $C(p) = 0$; region A includes this endpoint and its exact local certificate, with $g = 0$ and $\lambda = 1/16$. At $p = 2/3$, the region-C and region-H coefficients agree as in (15); all four transition corners are audited. If $p = 1$, then $W = 1$ almost everywhere and both sides are 1.

Theorem 7.1. *For the triangle–C_4 one-vertex amalgam H (Graph Atlas 126) and every symmetric measurable graphon W of edge density $p \in [1/2, 1]$,*

$$t(H,W) \geq p^2(2p-1)\big(p^3 + (1-p)^3\big).$$

8 Audit boundary and formalization status

The proof is graph-specific. It uses the factorization at the common vertex and a supporting plane tailored to three scalar rooted statistics. It does not assert a general vertex-amalgamation, cone, or join closure theorem. On the low interval, λ is calibrated sharply to the triangle-density profile; a coarser triangle bound does not prove the target.

The companion script also performs 46 exact two-block graphon checks. Each independently enumerates the homomorphism density, verifies the rooted factorization and scalar constraints, and checks the projection and target inequalities. They are sanity checks, not a replacement for the arbitrary-graphon argument.

No Lean command was run and no claim of Lean verification is made. A future formalization should define the graph in `lean/Taeyoung/Examples/Graph126.lean`, import or prove the sharp triangle profile, formalize the weighted Cauchy projection, and reflect the exact rational scalar certificates.

References

[1] Christian Reiher, *The clique density theorem*, Annals of Mathematics **184** (2016), 683–707.